

PROJECT KRATOS - ENC Inductions

Week 4, Q1

- Convention:-
- x-axis towards rover's forward heading line.
 - y-axis towards left of the rover (towards centre of rotⁿ)
 - Steering angle θ_i measured rel. to fwd. x-axis ($\hat{i}$), where ACW is +ve.

Inst. Centre of Rotⁿ $O = (x_0, y_0)$.

Wheel posⁿ = (x_i, y_i) .

Steering dirⁿ $\perp$ Line connecting O to wheel.

* Posⁿ vector:- $\vec{r}_i = (x_i - x_0)\hat{i} + (y_i - y_0)\hat{j}$

* $\perp$ Steering Angle (θ_i):- We know, $m_{\text{radius}} = \frac{y_i - y_0}{x_i - x_0}$.

Since wheel vel. vector $\vec{v}_i \perp \vec{r}_i$,

$$\begin{cases} m_{\text{radius}} \cdot m_{\text{steer}} = -1 \\ m_{\text{steer}} = -\frac{x_i - x_0}{y_i - y_0} \end{cases}$$

o.o Steering Angle $\theta_i \in [-90^\circ, +90^\circ]$ is given by:-

$$\theta_i = \tan^{-1}\left(-\frac{x_i - x_0}{y_i - y_0}\right)$$

Given $\rightarrow$ FL = (0.2, 0.2), FR = (0.2, -0.2), RL = (-0.2, 0.2), RR = (-0.2, -0.2)
[Wheel coordinates (x_i, y_i)]

Ans

WHEEL

Case I: $O = (0, 0)$

Case II: $O = (1, 0)$

Case III: $O = (-2, 0)$

Front Left (FL)

-45.00°

+75.96°

+84.81°

Front Right (FR)

+45.00°

-75.96°

+84.81°

Rear Left (RL)

+45.00°

+80.54°

-83.66°

Rear Right (RR)

-45.00°

-80.54°

+83.66°

*NOTE:- The wheel radius ($r = 0.08m$) is req^d for calculating individual wheel rotⁿ speeds but does not affect steering dirⁿ (θ_i) of wheels.

$$\omega_i = \frac{v_i}{r_{\text{wheel}}} = \frac{v_{\text{rover}} \cdot d_i}{r_{\text{wheel}}}$$

(d_i = dist. (radius) from centre to wheel)